Credit modeling: defensible evidence

A reproducible portfolio audit of logistic regression and XGBoost

Signal, uncertainty and failure analysis

Data audit → controlled ablations → nested development CV → calibration → interpretation

Historical test repeatedly inspected; excluded from every new experiment.

The outcome is a stronger study, not a score claim

Both families have modest development signal. The evidence also shows overfitting and costly operating points.

GLASSBOX SELECTOR AP

0.131

Fold SD 0.033

BLACKBOX SELECTOR AP

0.131

Fold SD 0.010

DUMMY AP

0.061

Natural-prevalence reference

- Outlier handling matters; additional feature removal is mixed.
- Separate calibration repairs probability levels. Neither family has proven universal superiority.

Data provenance remains the main deployment blocker

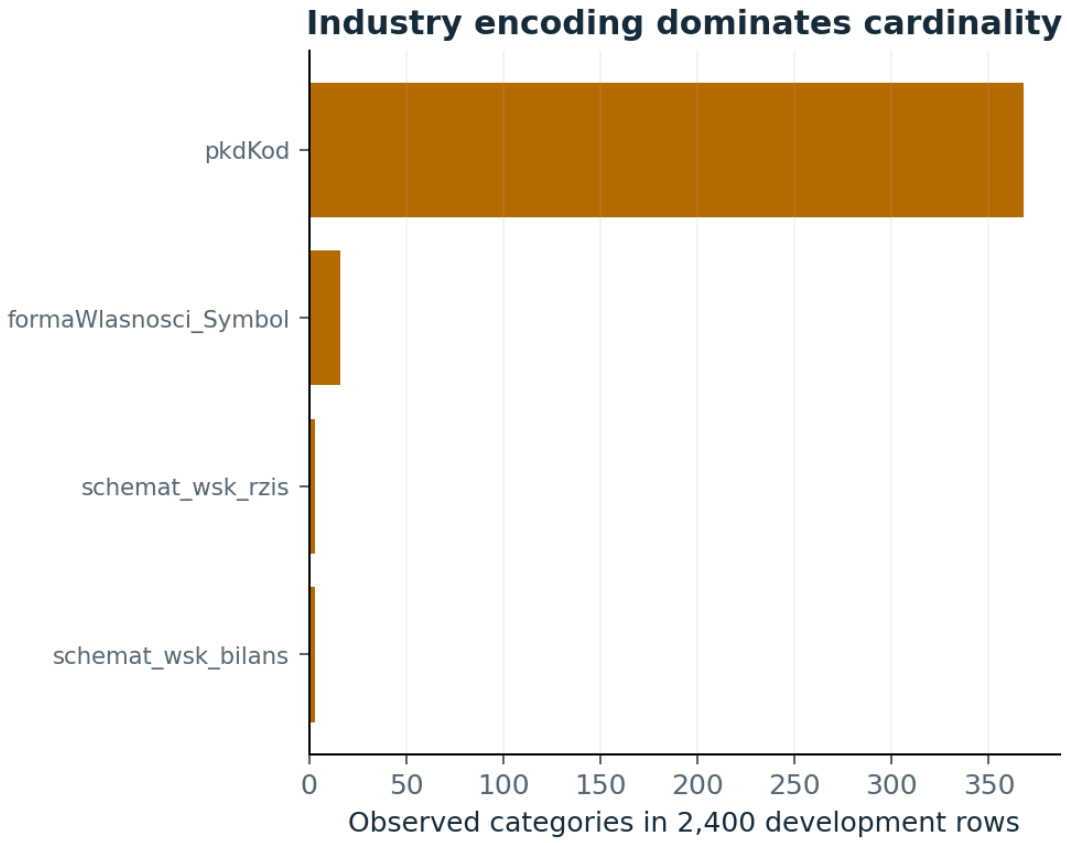
Binary labels are observed; a default horizon, decision dates and borrower identities are not documented.

Verified	Unresolved
3,000 rows / 183 defaults	Default event definition and horizon
218 configured raw predictors	Feature availability at decision time
No exact duplicate records	Repeated borrowers cannot be excluded
Historical 600-row exclusion frozen	No new external or temporal cohort
2,400 development rows / 146 defaults	No verified timestamps for temporal validation

The audit found concrete quality problems

These are measured problems; near-zero denominators and post-outcome leakage remain hypotheses.





Redundancy is real, but removal needs evidence

Eight exact duplicate field pairs; 66 numerical pairs with |Spearman rho| ≥ .98 on development data.

- Example exact duplicates: wsk_ROS / wsk_rent_sprzedazy; wsk_koszty_przychody / wsk_poziom_kosztow_operacyjnych.
- Some ratios reach about 10^17; 1,014 infinite cells occur in the supplied full CSV.
- Two coincident four-field profiles are not verified borrower IDs. No complete borrower-aware split is supported.
- Every raw field’s prediction-time availability remains explicitly unverified.

Historical evidence showed overfit before the test

Old training, validation and selected-CV results are diagnosis only; they do not select new models.

Historical configuration	Train AP	Validation AP	Selected CV AP
Logistic baseline	0.611	0.106	—
XGBoost baseline	0.991	0.186	—
Balanced L1 selection	0.298	0.140	0.137
Heavily regularized XGBoost	0.230	0.172	0.157

The old tree winner reduced variance and fitted signal. Investigate regularization and sampling, not test-tune.

Repeated nested development CV

Two fixed repeats (42, 43) × three outer folds; three-fold inner AP selection.

960 rows

Fit and inner search

≈58–59 defaults

320 rows

Independent sigmoid

≈19–20 defaults

320 rows

Independent F2 threshold

≈19–20 defaults

800 rows

Outer development assessment

≈48–49 defaults

All four roles are disjoint. This costs fitting data, but avoids sharing calibration and threshold labels.

Random stratification is a documented limitation, not a substitute for missing borrower/time metadata.

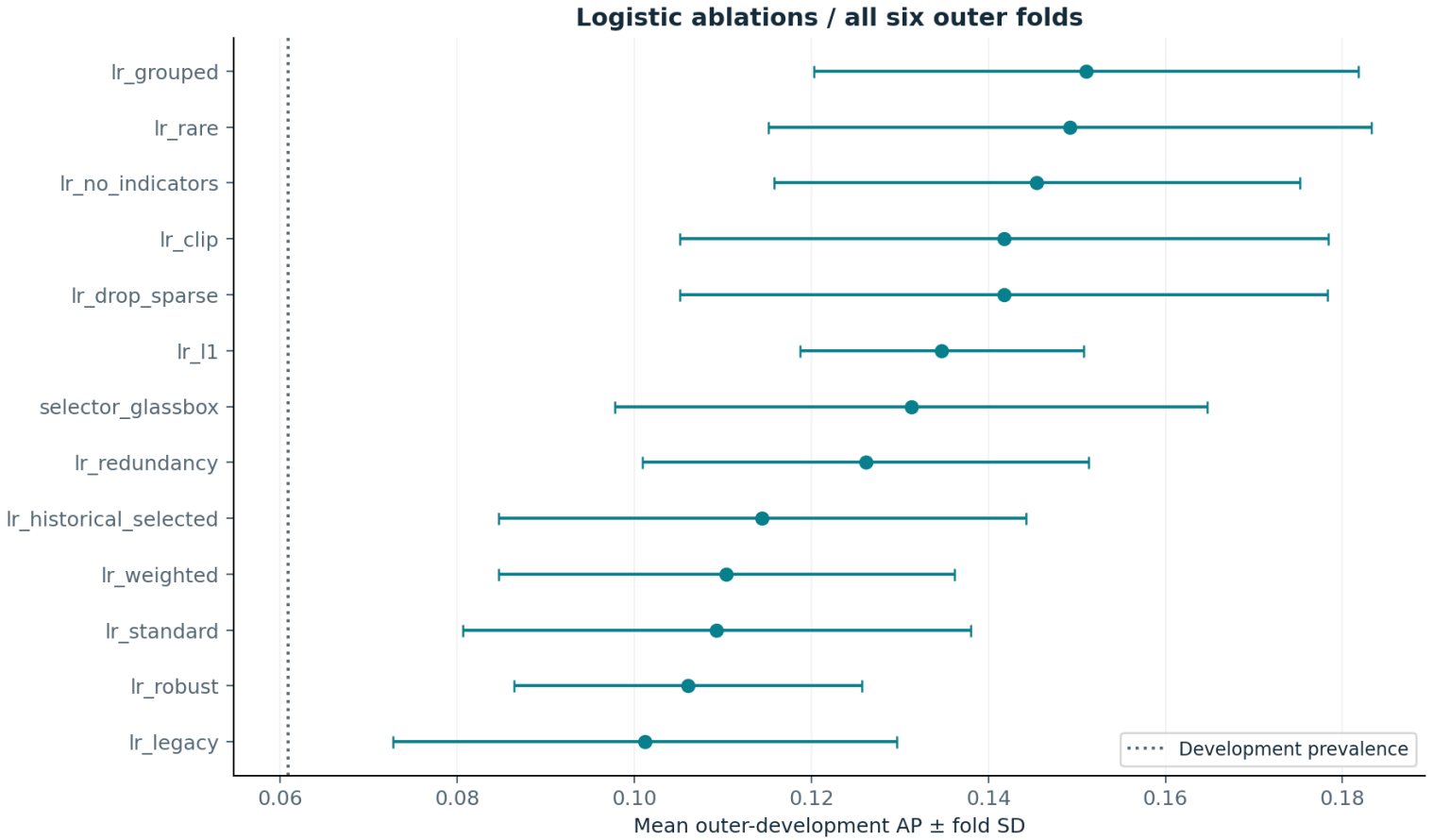
A small set of explicit experiment families

All ablations use the same outer roles; feature settings are selected inside inner folds.

Question	Controlled comparison
Imbalance	Unweighted vs balanced logistic; identical tree candidates with fold weights
Extreme values / scaling	1%/99% train-only clipping; robust vs standard scaling after clipping
Missing / rare data	Indicators off; sparse-field cutoffs; rare-level pooling
Feature selection	L1; raw-field Spearman redundancy; whole-field supervised top-k
Tree search	Matched rate/tree-count paths, moderate complexity controls, frozen references

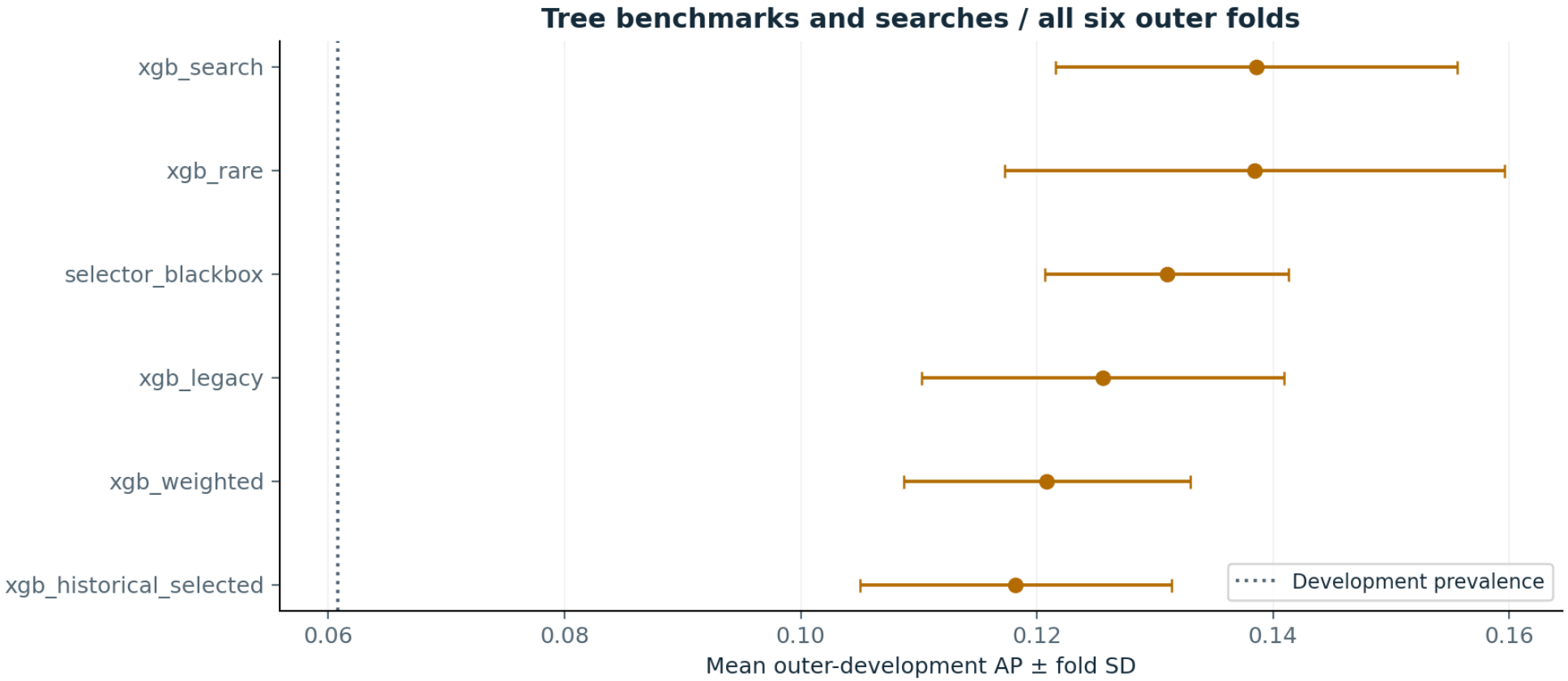
The full logistic ablation table is retained

Mean ± SD over six outer folds. SD is descriptive; these folds are dependent.



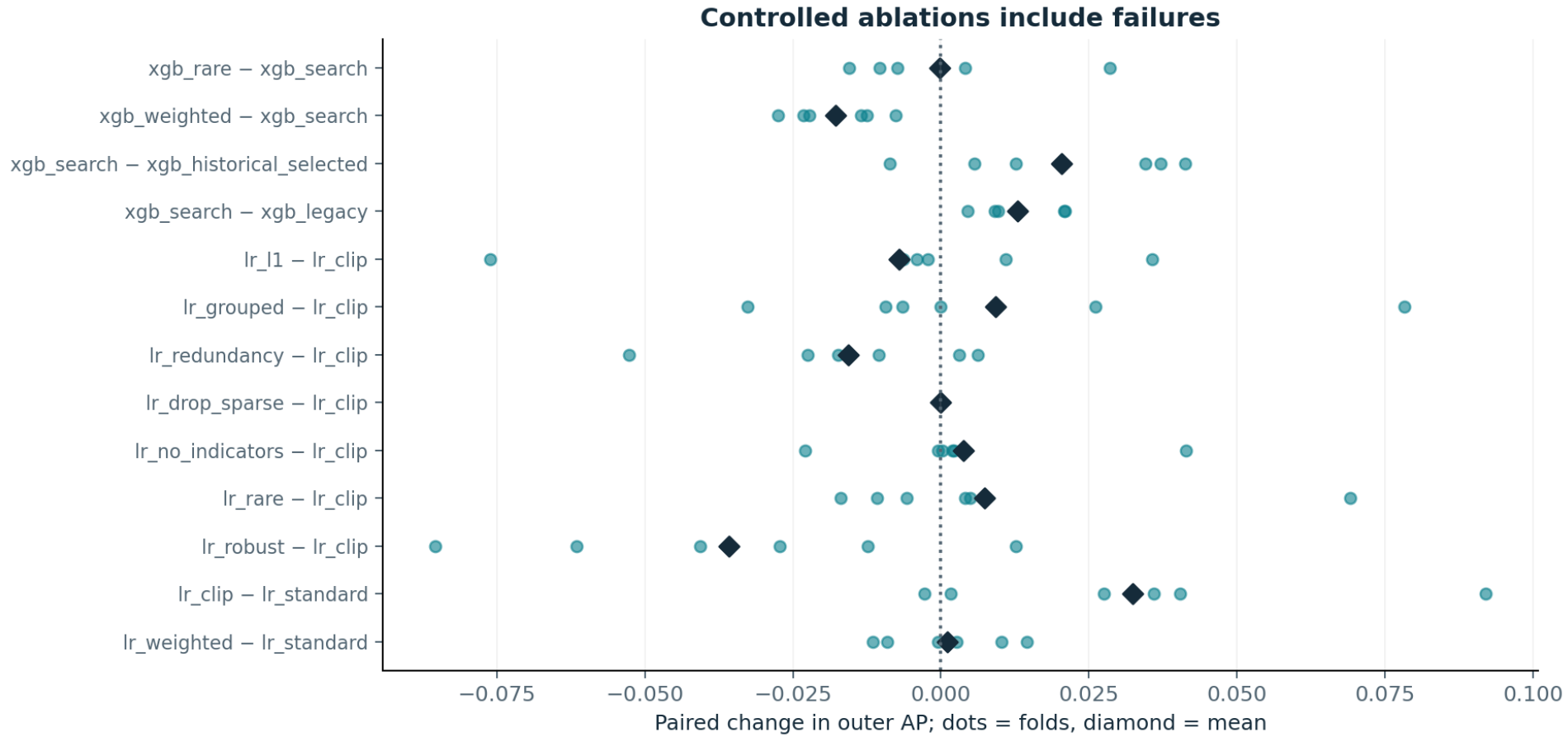
Trees: moderate controls, matched search paths

No early stopping: tree counts are inner-selected. Weighted and unweighted searches share eight candidates.



Paired changes include negative findings

Dots show all six paired outer-fold AP changes; diamonds are means. No favorable seed was selected.



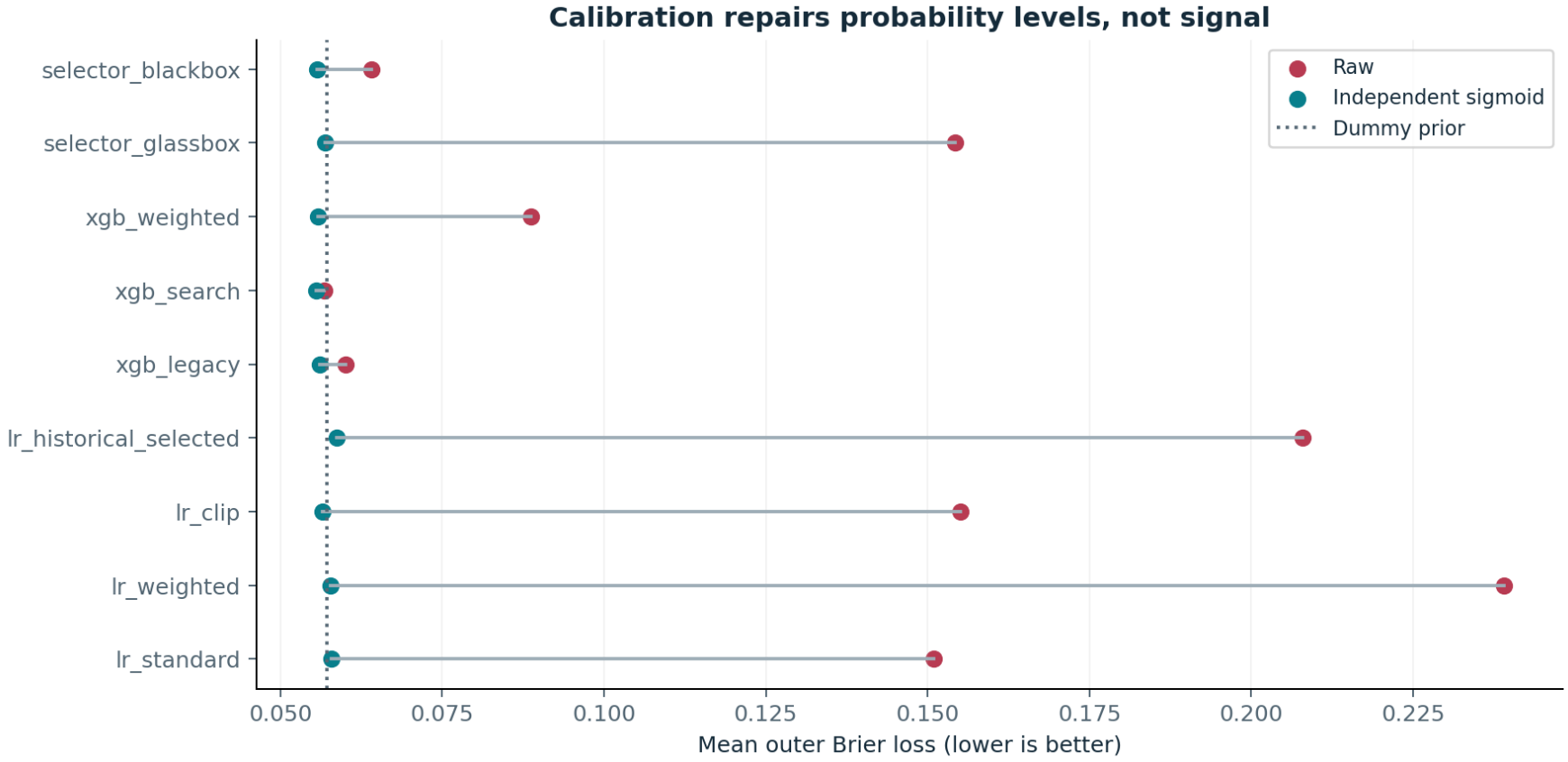
Feature selection is not automatically an improvement

Whole raw fields preserve one-hot groups; dimensions and active sets vary across training folds.

Recipe	Raw fields	Dimensions	Active coeff.	AP
lr_clip	218.0	494.5	494.5	0.142
lr_rare	218.0	251.5	251.5	0.149
lr_drop_sparse	215.0	488.5	488.5	0.142
lr_redundancy	160.7	435.3	435.3	0.126
lr_grouped	91.7	145.0	145.0	0.151
lr_l1	218.0	494.5	22.5	0.135

Calibration is a separate probability experiment

A regularized monotone sigmoid fits natural-prevalence rows independently of model fitting and thresholds.



A fitting diagnosis: the intercept was penalized

Adaptive follow-up, reported separately. It uses the same folds and C grids; it is not independent confirmation.

- liblinear penalizes its synthetic intercept. Very small C can pull even an unweighted score toward 0.5.
- L2 follow-up: lbfgs with an unpenalized intercept, with and without clipping.
- L1 follow-up: same solver, with intercept_scaling=100 to weaken the intercept penalty.

Clipped L2 raw Brier: liblinear 0.155 → lbfgs 0.057.

Primary procedure-level results

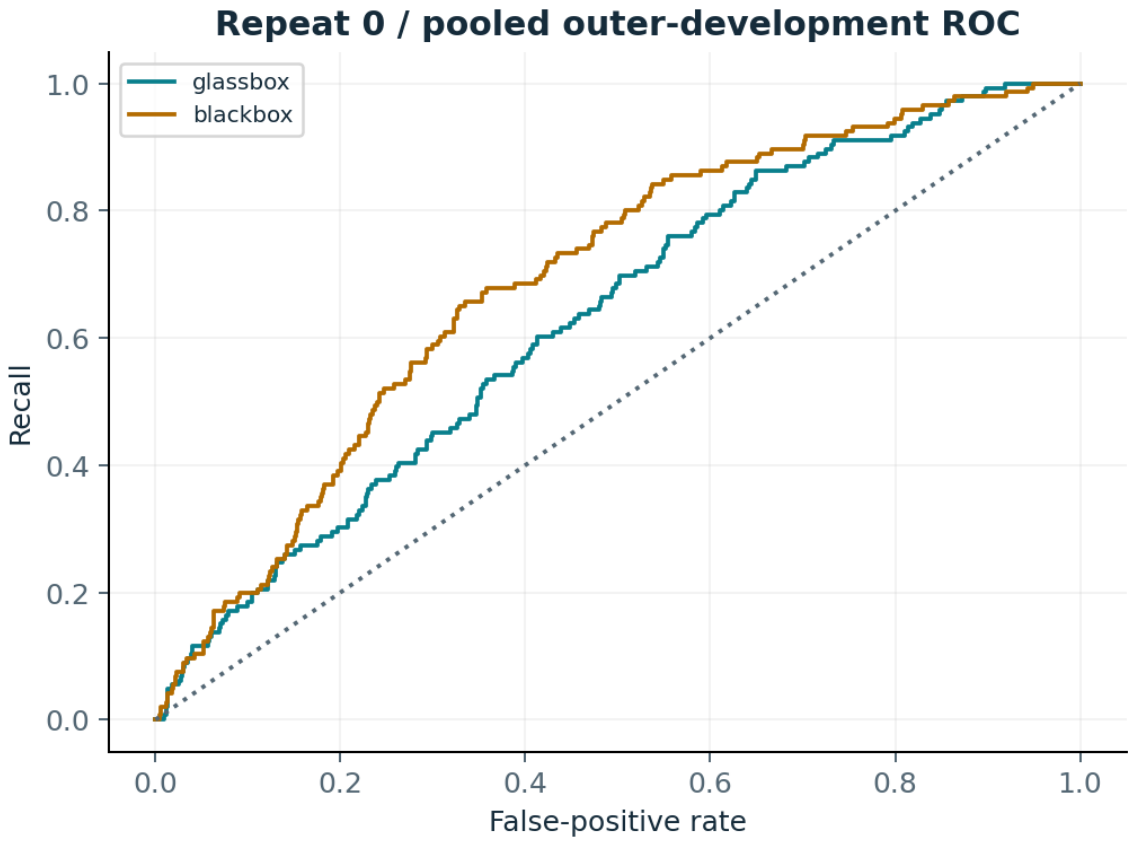
Each family chooses its recipe by inner AP only. These are development estimates, not a new test result.

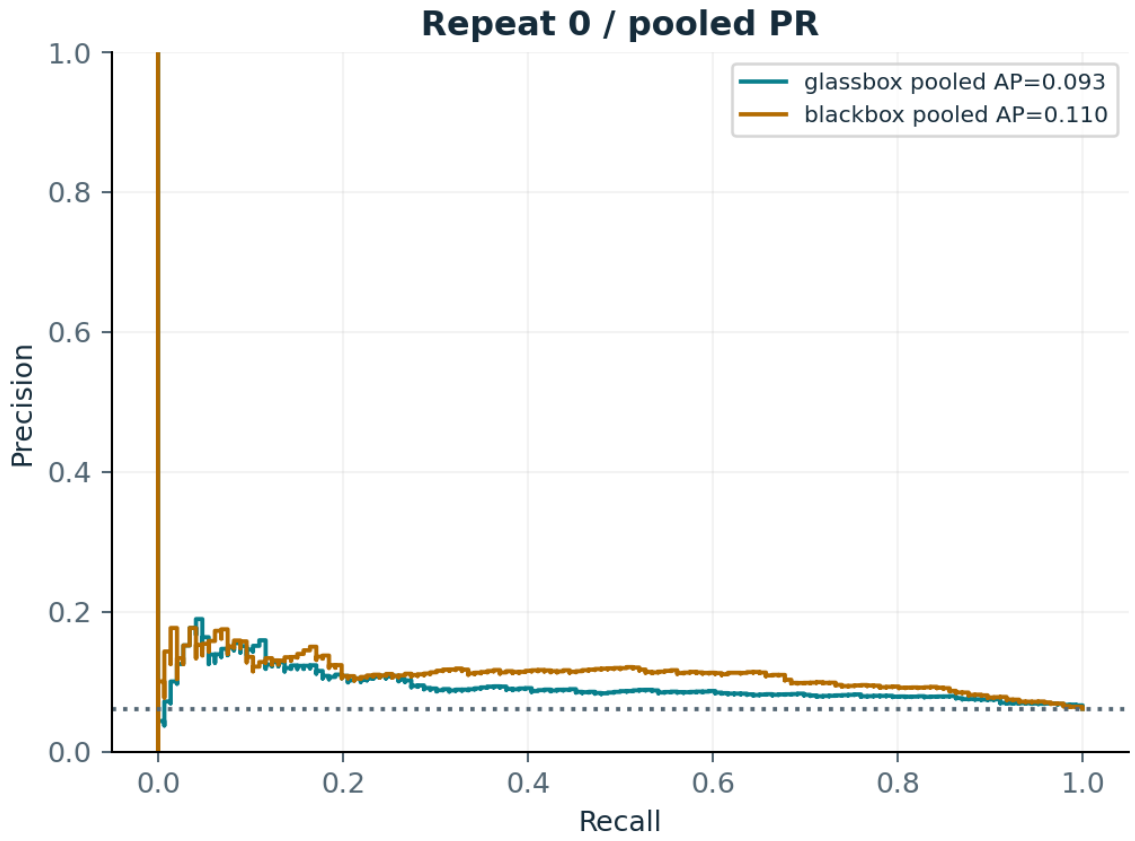
Procedure	AP ± SD	ROC-AUC	Brier	Log loss
dummy	0.061 ± 0.001	0.500	0.0571	0.229
glassbox	0.131 ± 0.033	0.659	0.0570	0.226
blackbox	0.131 ± 0.010	0.706	0.0556	0.215

Sigmoid outputs shown. The adaptive intercept study is separate; it does not replace these planned estimates.

Ranking curves need scope

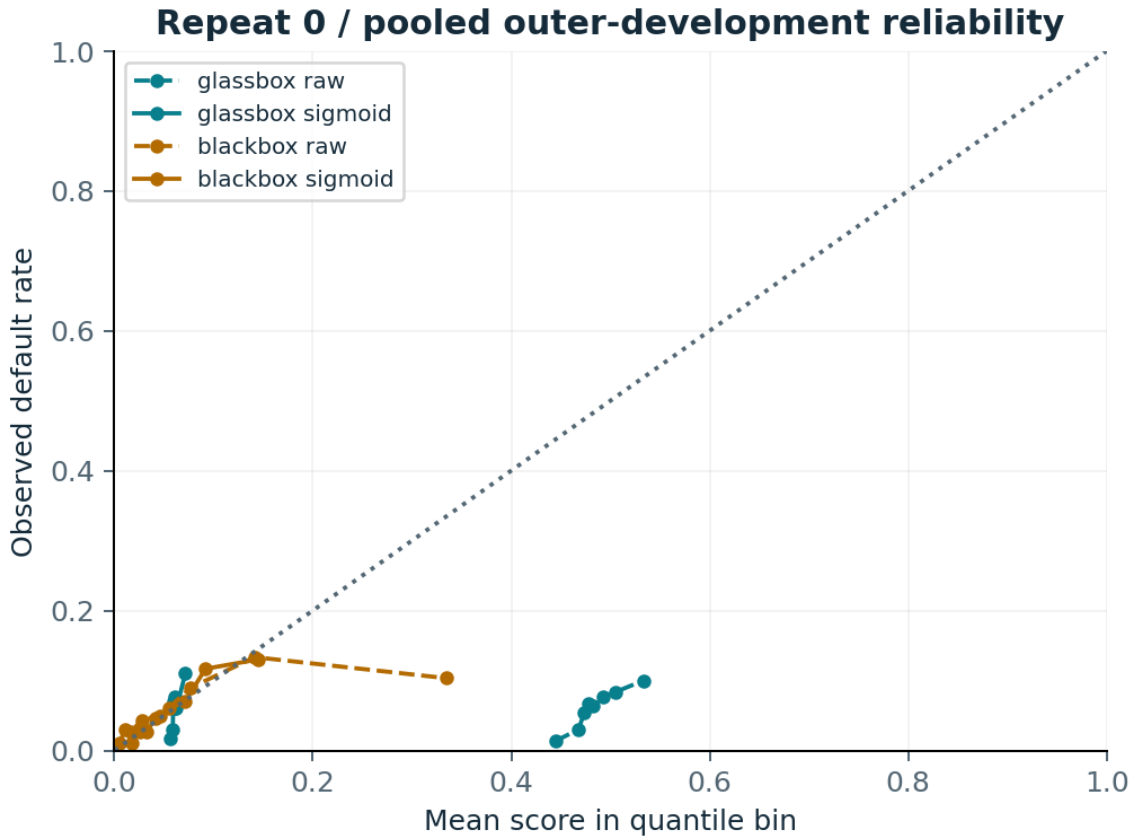
Repeat 0 pooled outer-development scores. Pooled AP and mean fold AP are different summaries.

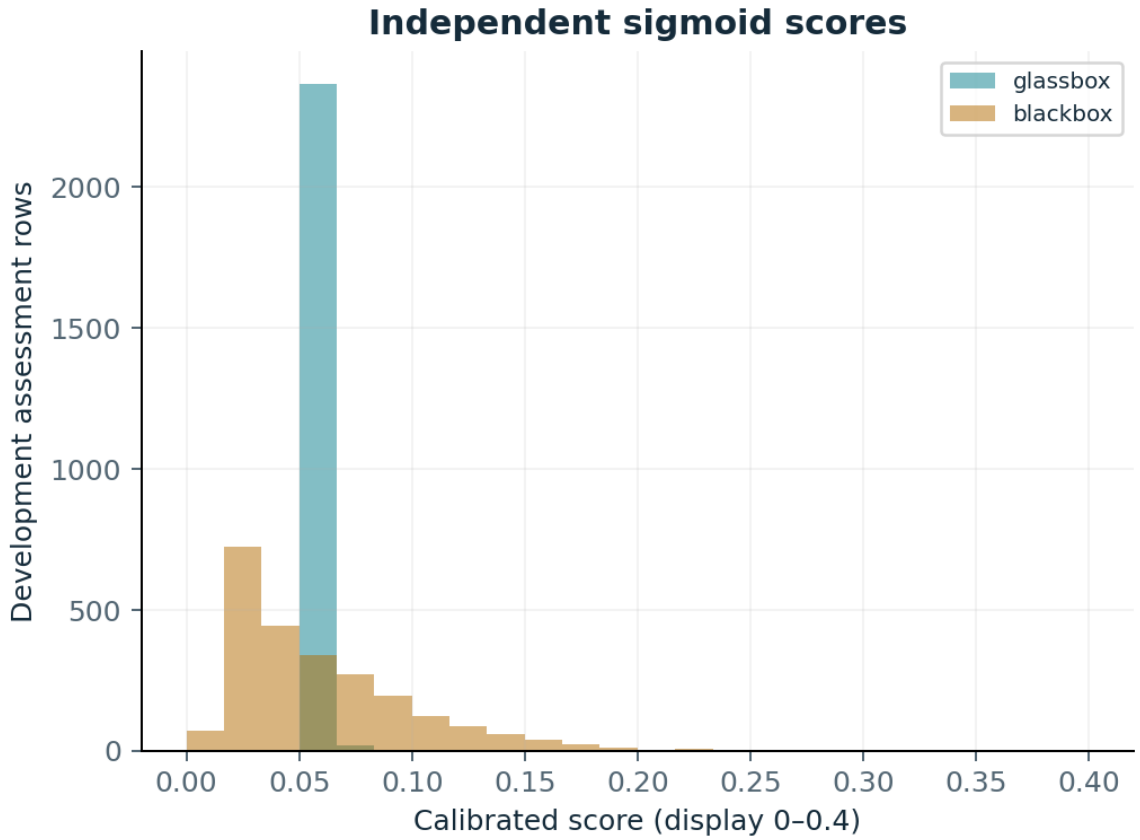




Probability reliability remains uncertain

Repeat 0 pooled outer-development predictions; the fitted calibration subsets contain only ≈20 defaults each.





F2 thresholds still create substantial workload

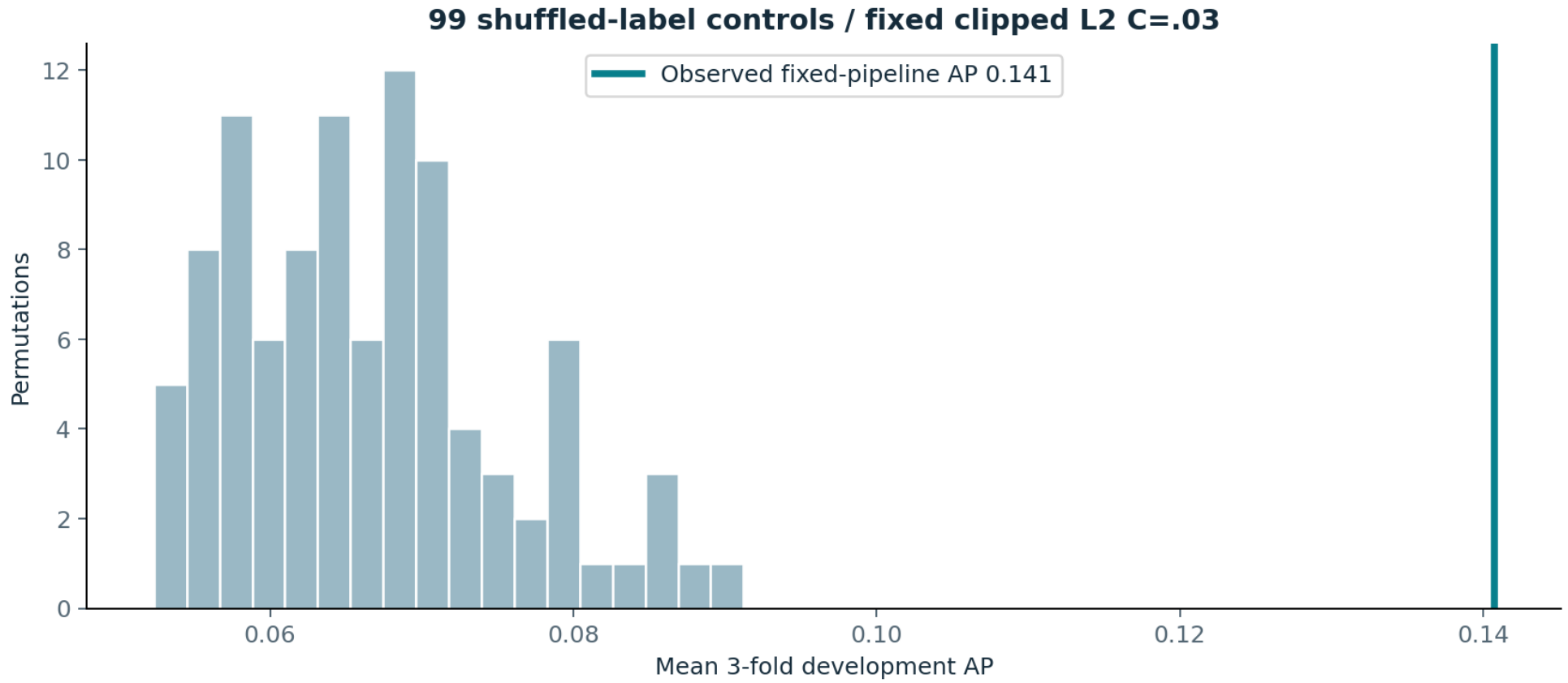
F2 is a stated recall preference, not a monetary cost model. Values are outer-fold means.

Procedure	Precision	Recall	FP / 800	Flagged
dummy	6.1%	100.0%	751.3	100.0%
glassbox	8.4%	67.0%	362.0	49.3%
blackbox	10.2%	65.4%	290.5	40.3%
lr_lbfgs_clip	10.1%	54.0%	238.8	33.1%

Last row is the adaptive follow-up. No threshold was selected from the historical test or outer labels.

There is signal; no ceiling is established

Fixed clipped L2 C=.03, three-fold development CV, 99 shuffled-label controls.



Uncertainty rules out a strong winner claim

Original rows are paired across repeats in the bootstrap; model refitting is not included.

CONDITIONAL ΔAP INTERVAL

[-0.033, +0.033]

Tree selector minus logistic selector

- Fold SD, repeat-level results and paired ablations are all retained.
- The interval conditions on fitted predictions and prevalence; unknown borrower clustering is not modeled.
- Adaptive experimentation and future distribution shift require a new external / temporal cohort.

Both families: strengths and costs

Interpretability does not compensate for poor probability estimates or incomplete data provenance.

	Logistic regression	XGBoost
Strength	Signed additive margins; sparse variants	Nonlinearities and interactions
Weakness	Representation and extreme-value sensitivity	Overfit and tuning complexity
Probability issue	Weighting and intercept penalization	Small calibration sets; score shrinkage
Interpretation	Exact native terms; Linear SHAP	Gain has no direction; Tree SHAP
Observed limit	No clear family-level AP win	Low LIME fidelity in the studied model

Expanded interpretation, without favorable-case selection

Predeclared outer repeat 0/fold 0; recipe chosen by inner AP. No historical test explanations are added.

SHAP ROWS

800 / family

All held-out rows in that outer fold

SUCCESS / ERROR CASES

32 / family

8 random TP, FP, FN, TN cases

LIME FITS

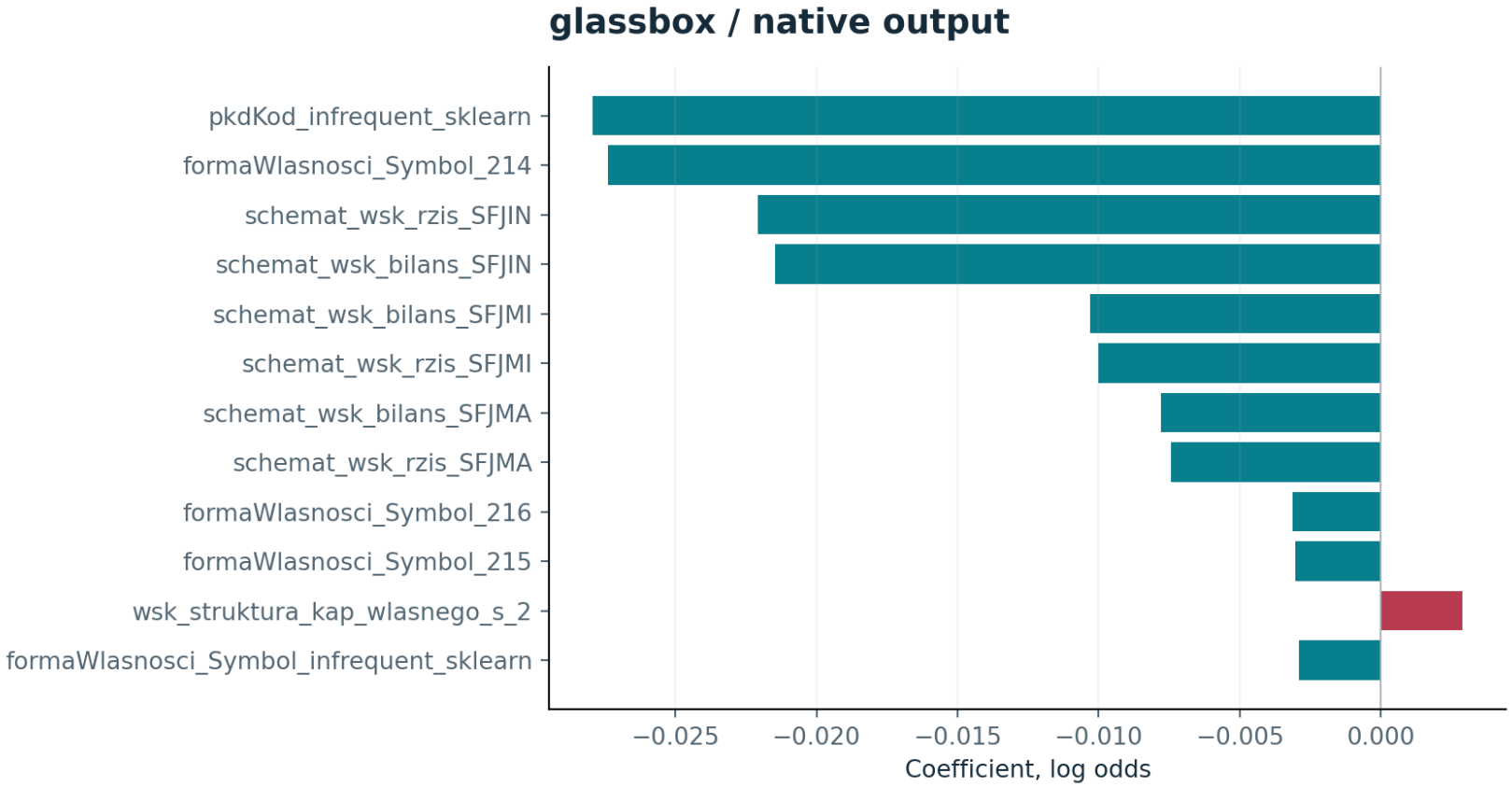
192 total

3 seeds × 32 cases × 2 families

- Three fit-only SHAP backgrounds test reference sensitivity.
- Native/SHAP explain base log odds. LIME approximates base probability; calibrated decisions are separate.

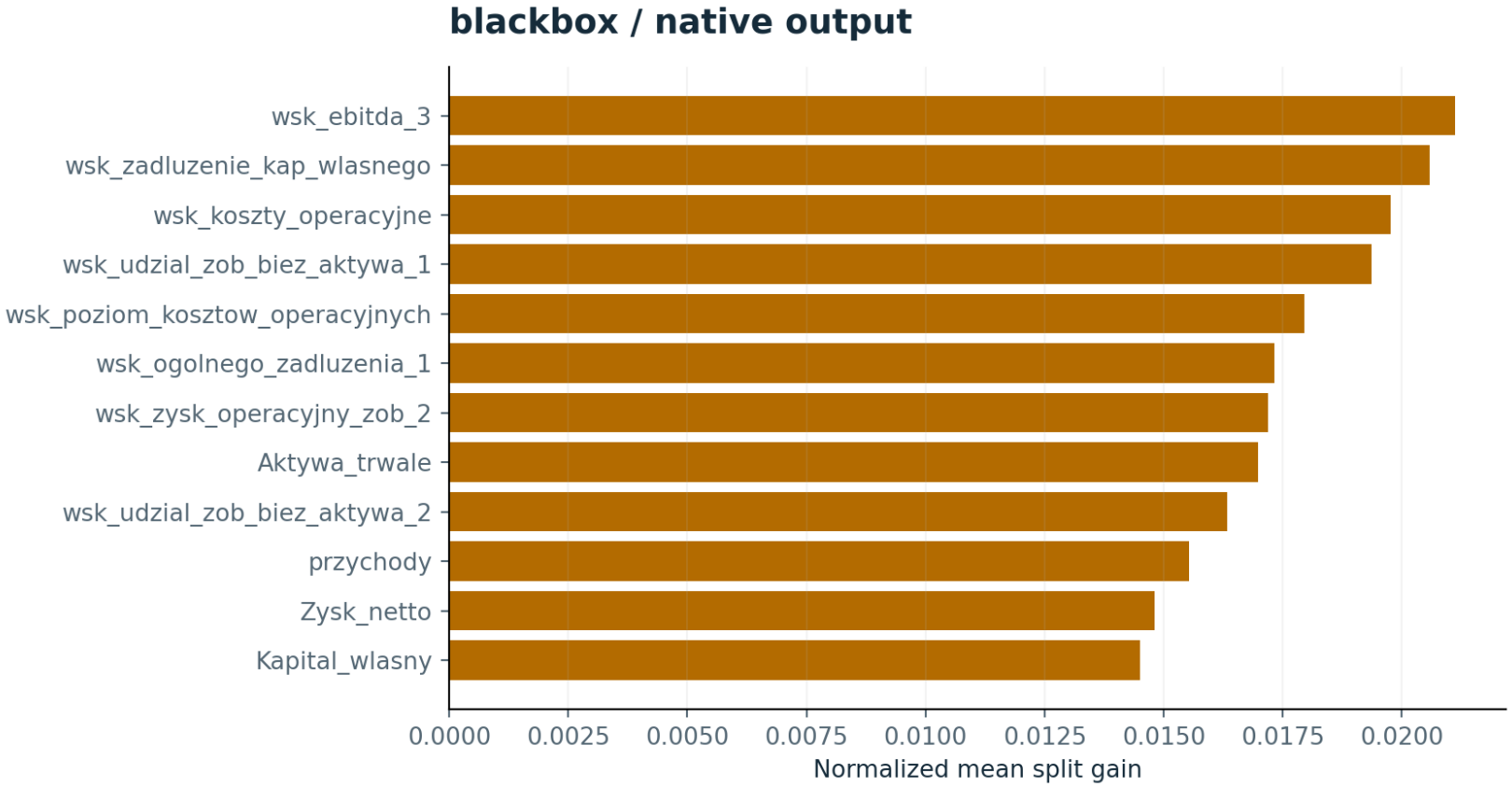
Native glassbox: actual fitted output

Coefficient signs describe the base logistic margin; tree gain is unsigned training split importance.



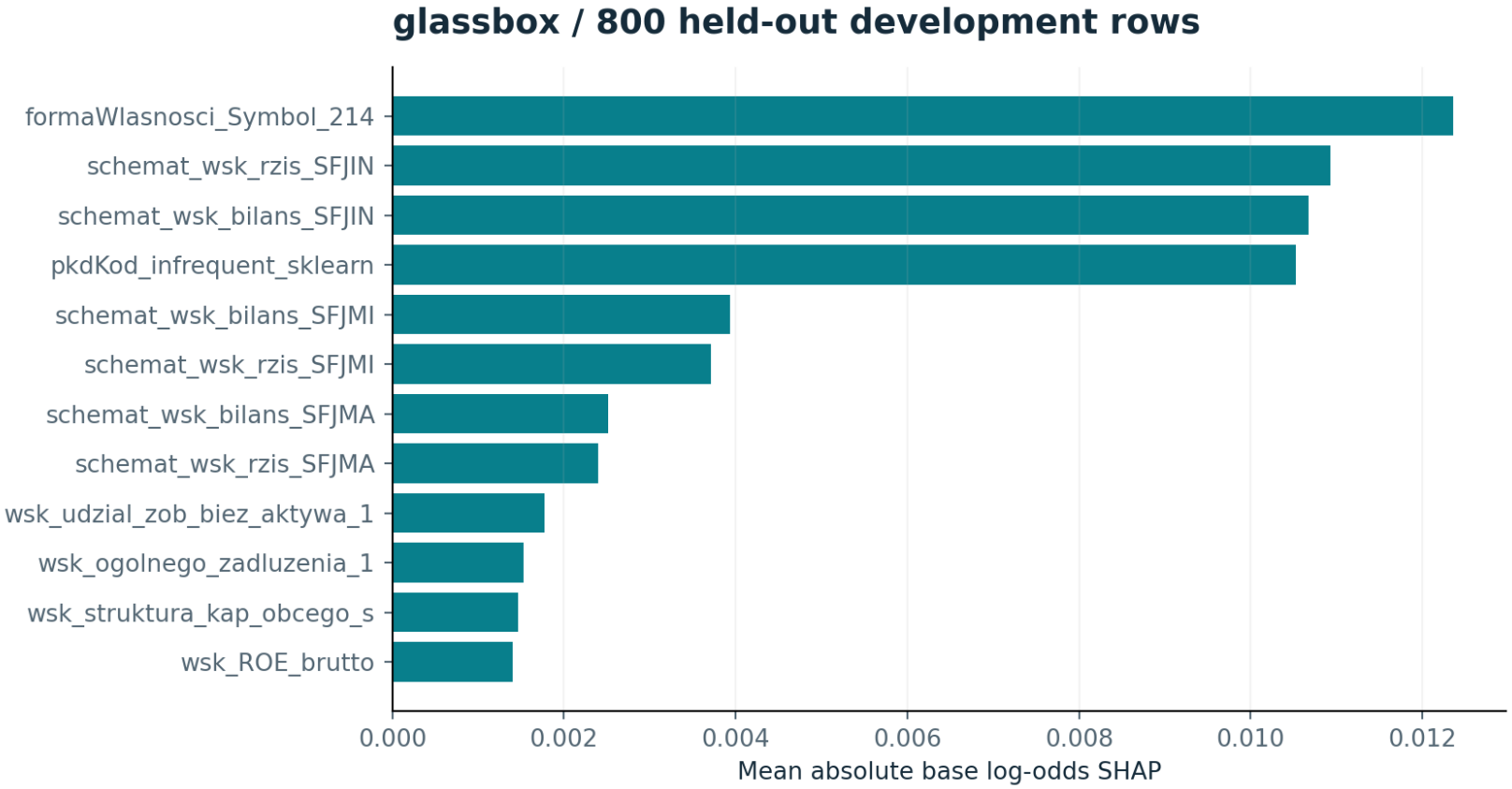
Native blackbox: actual fitted output

Coefficient signs describe the base logistic margin; tree gain is unsigned training split importance.



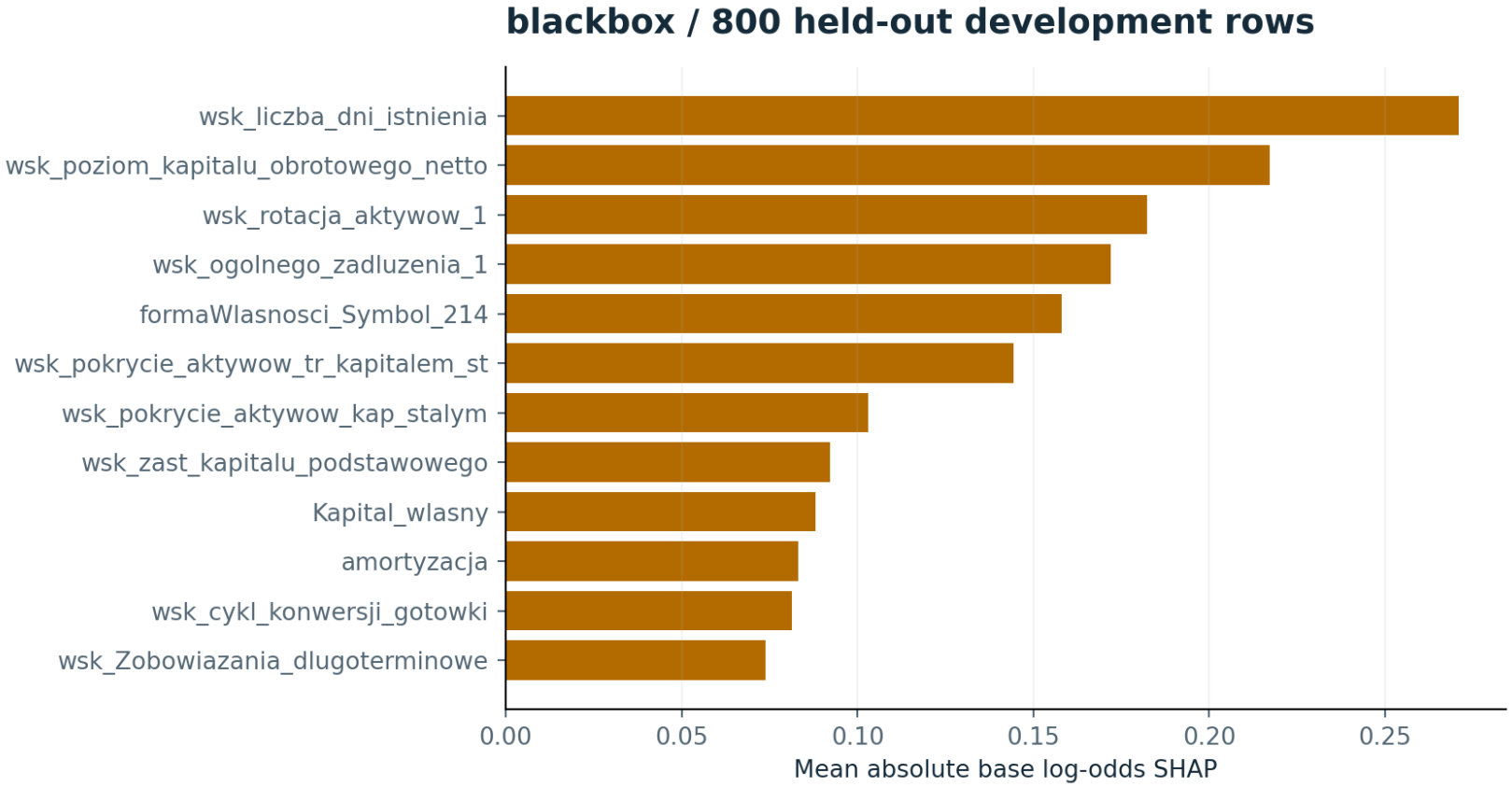
Global SHAP: glassbox, 800 development rows

Mean absolute base-log-odds attribution averaged across three training backgrounds.



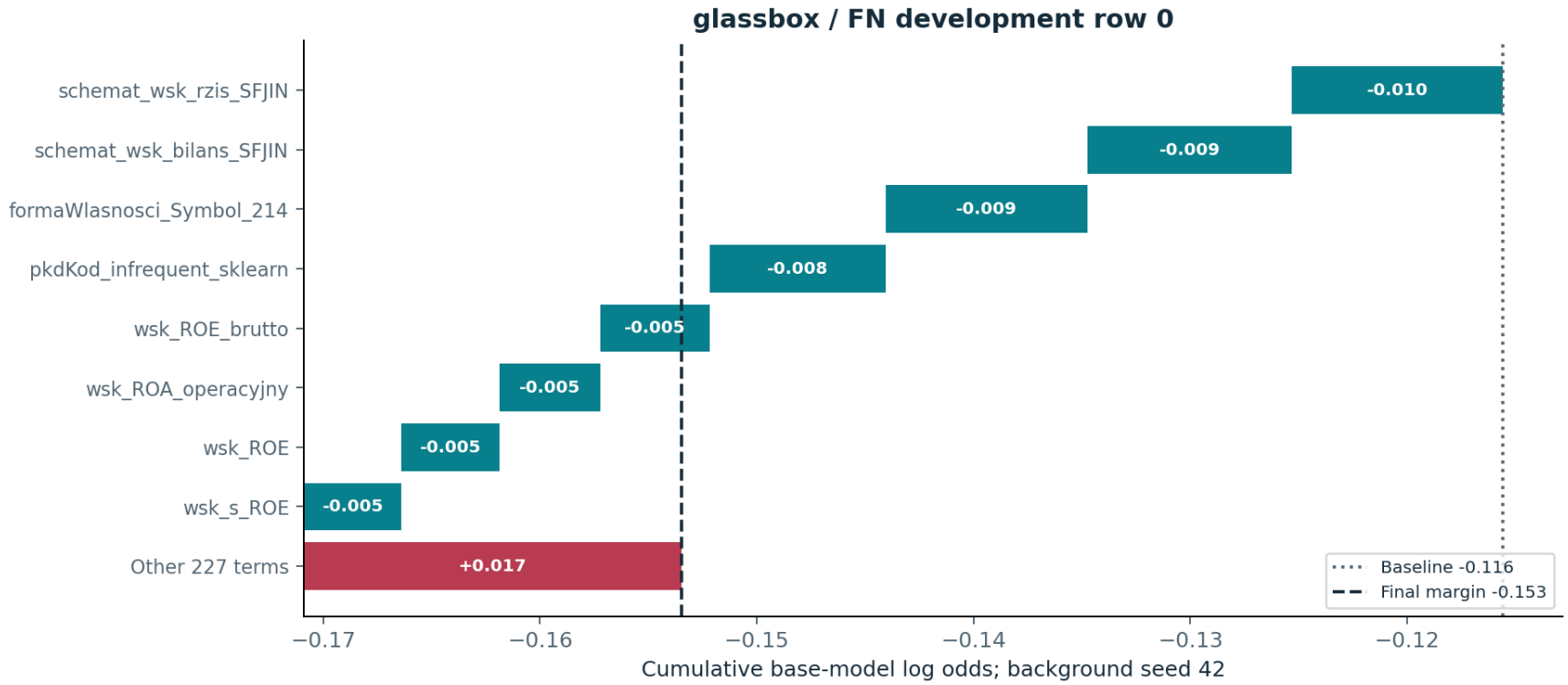
Global SHAP: blackbox, 800 development rows

Mean absolute base-log-odds attribution averaged across three training backgrounds.



Error analysis: glassbox, a missed default

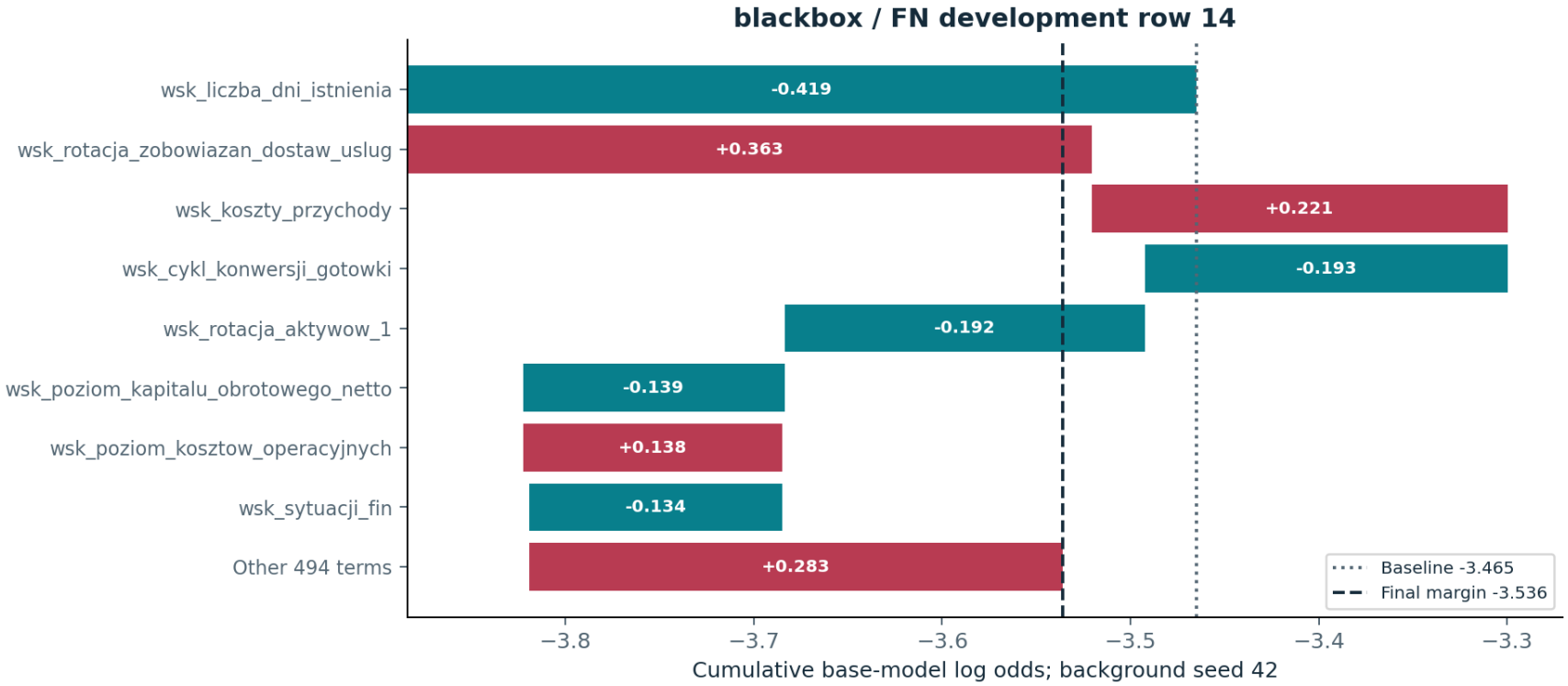
First randomly sampled FN in the fixed development fold; signed effects explain the base classifier.



Row 0: actual 1; base 0.4617 → sigmoid 0.0612 < t 0.0627.

Error analysis: blackbox, a missed default

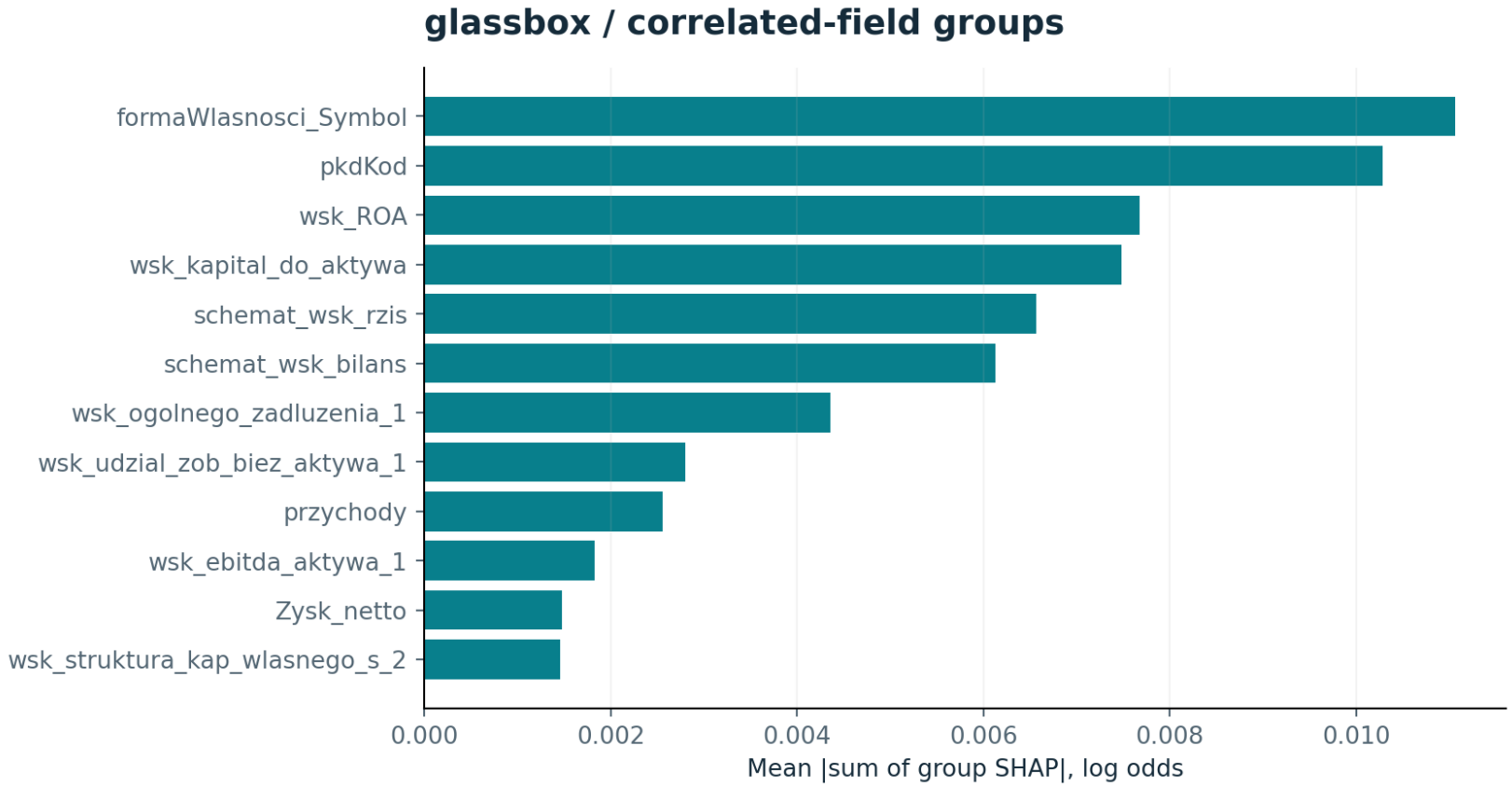
First randomly sampled FN in the fixed development fold; signed effects explain the base classifier.



Row 14: actual 1; base 0.0283 → sigmoid 0.0443 < t 0.0541.

Correlated fields can redistribute credit

Fit-only correlation groups sum signed SHAP terms before taking mean absolute magnitude.



Explanation stability has several meanings

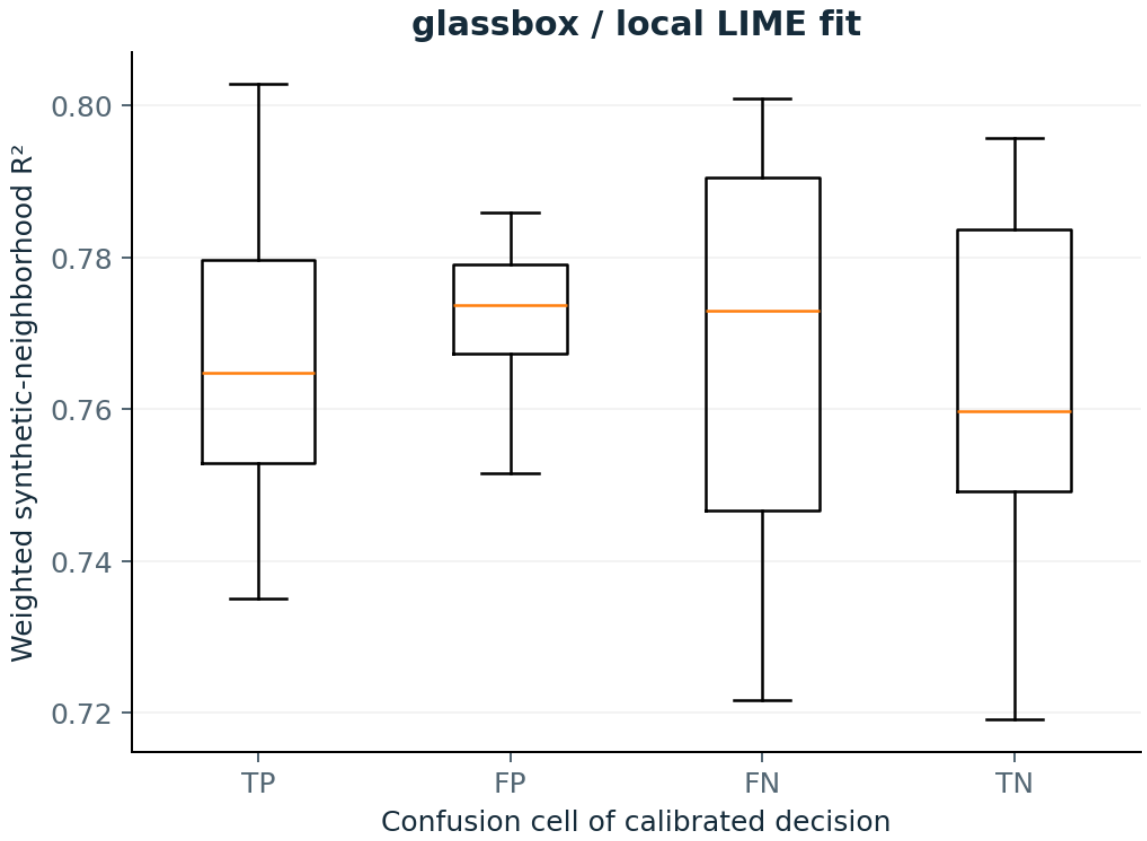
Background stability does not establish retraining stability, causal validity or subgroup fairness.

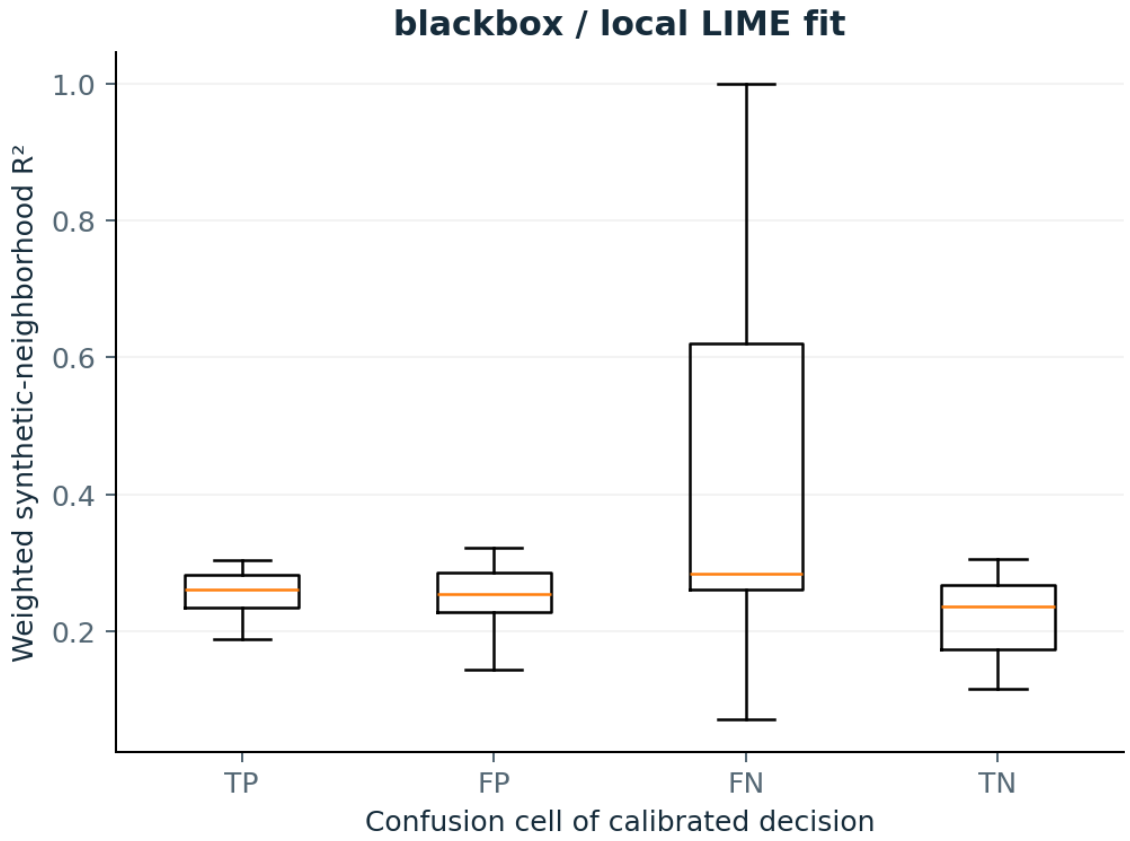
Family	SHAP top-10 Jaccard	SHAP rank correlation	LIME selected-15 Jaccard
glassbox	1.000	0.970	0.984
blackbox	0.879	1.000	0.587

Active/retained raw-field Jaccard across all six fitted folds is also exported for every recipe.

LIME fidelity is weak for this tree model

32 cases/family × 3 seeds × 2,000 perturbations; synthetic-neighborhood R^2 is not held-out faithfulness.





Successes and errors are all retained

Balanced confusion-cell sampling is illustrative; it is not a population-weighted case distribution.

Family	Cell	Cases	Median base	Median sigmoid
blackbox	FN	8	0.025	0.041
blackbox	FP	8	0.065	0.074
blackbox	TN	8	0.015	0.029
blackbox	TP	8	0.115	0.107
glassbox	FN	8	0.470	0.062
glassbox	FP	8	0.476	0.063
glassbox	TN	8	0.466	0.062
glassbox	TP	8	0.481	0.064

Reproducibility is part of the result

Every new saved model and every role boundary was checked; historical artifacts were preserved.

- 126 saved base models: assessment predictions and metrics recomputed after loading.
- 19 tests, including outer-label isolation and portable historical exclusions.
- Refit checks are distinct from saved-model checks; inner L1 search scores vary slightly between runs.
- Published code, plan, fold tables, runtimes, environment and presentation; original run snapshots and theory PDF stay local.

What would make this deployment evidence?

Implemented: rigorous local development study. Proposed: external validation and operational design.

- Obtain borrower identifiers, dates, target horizon and verified prediction-time feature lineage.
- Use appropriate group/time development splits, then assess a genuinely new cohort.
- Define error costs, review capacity, subgroup checks and monitoring ownership.
- Validate recalibration and explanation stability under refitting and distribution shift.

Evidence and methods

Technical report, theory guide and compact machine-readable evidence accompany this presentation.

- docs/portfolio-audit.md and docs/evidence/portfolio_v1: complete evidence and limitations.
- config/portfolio.yaml: planned study; portfolio_intercept.yaml: labeled adaptive follow-up.
- Primary sources: scikit-learn CV/scoring/calibration; XGBoost; SHAP (Lundberg & Lee); LIME (Ribeiro et al.).
- GitHub reports/portfolio_v1: presentation PDF and PowerPoint. Theory PDF, technical-report PDF, charts and previews stay local.